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EDUCATION

Guangzhou Maritime University

09/2022-07/2026

B. Eng. in *Engineering Management* | School of Future Transportation

Thesis: Attribution Analysis of Digital Retrofitting Effectiveness in Urban Transportation Infrastructure

Research Interests: Mixed-Autonomy Systems; Traffic Flow Modeling and Control; AI for Transportation

ACADEMIC VISITS

Massachusetts Institute of Technology

05/2026-10/2026

Selected as one of 41 global participants in *MIT-UF-NEU Summer Research Camp* at Prof. Cathy Wu's Lab, collaborating with Ryan Hardesty Lewis on distilling interaction preferences for mixed-autonomy driving.

Tsinghua University & Technical University of Munich

12/2025

Selected as one of 67 global participants for the "*Smart Sustainable Cities and Transportation*" Winter School, completing 11 modules covering data-driven modeling, mixed-traffic systems, and urban computing.

CONFERENCE PAPERS

[C1] **Deng, H.**, He, S., Wang, D. (2026). FLAT: Smoothing the Rugged Landscape for Learnable, Sample-Efficient Traffic Calibration. Preparing for submission to *The 38th IEEE Intelligent Vehicles Symposium (IV)*. DOI: [10.48550/arXiv.2610.04337](https://doi.org/10.48550/arXiv.2610.04337).

[C2] **Deng, H.**, Zheng, F., Xia, X. (2026). SpinFlow: A Physics-Informed Spin Field Framework for Traffic Phase Inference and Transition Detection. Accepted at *The 29th IEEE International Conference on Intelligent Transportation Systems (ITSC)*. (**Oral Presentation**). DOI: [10.48550/arXiv.2605.23306](https://doi.org/10.48550/arXiv.2605.23306).

[C3] **Deng, H.**, Zheng, F., Ma, K., et al. (2025). Adaptive Truncated Schatten Norm for Traffic Data Imputation with Complex Missing Patterns. *The 25th COTA International Conference of Transportation Professionals (CICTP)*. American Society of Civil Engineers. (**Best Poster Award**). DOI: [10.1061/9780784486269.009](https://doi.org/10.1061/9780784486269.009).

[C4] Xia, X., **Deng, H.***, Ye, Y. (2026). Fundamental Diagram Modeling and Simulation of Mixed Traffic Flow Considering Functional Degradation. Accepted at *The 26th CICTP Conference*. *Corresponding author.

JOURNAL ARTICLES

[J1] Xia, X., **Deng, H.**, Tang, J., et al. (2025). Dynamic Traffic Assignment under Mixed Traffic: Modeling, Evolution, and Solution Approaches. *Journal of Transport Science and Engineering*. (**Cover Paper, Presented orally at AIAT 2026**). (Chinese). DOI: [10.16544/J.CNKI.CN43-1494/U.20251007001](https://doi.org/10.16544/J.CNKI.CN43-1494/U.20251007001).

[J2] **Deng, H.**, Ma, K., Zhan, L., et al. (2025). Structural Equation Modeling of the Relationships among Traffic Safety KAP in Small-Medium Cities. *Journal of Transportation Engineering*. (Chinese). DOI: [10.13986/J.CNKI.JOTE.2026.04.002](https://doi.org/10.13986/J.CNKI.JOTE.2026.04.002).

[J3] **Deng, H.**, Tang, J., Zhan, L., et al. (2025). Research on Traffic Governance Optimization in Kaiping City Based on a GRA-FCE Integrated Framework. *Journal of Transportation Engineering*. (Chinese). DOI: [10.13986/J.CNKI.JOTE.2026.04.011](https://doi.org/10.13986/J.CNKI.JOTE.2026.04.011).

[J4] Wang, Y., Li, L., **Deng, H.**, et al. (2026). Sparse Stochastic Optimal Control under Control-Dependent Diffusion and Time-Varying Dynamics. *Journal of Dynamical and Control Systems*. DOI: [10.1007/s10883-026-09784-1](https://doi.org/10.1007/s10883-026-09784-1).

RESEARCH EXPERIENCE

Massachusetts Institute of Technology

05/2026-10/2026

Title: *Inference-Time Behavior Shaping of Autoregressive Driving Priors for Mixed-Autonomy Traffic Control* | Research Participant | PI: Prof. Cathy Wu | Mentor: Ryan Hardesty Lewis

- Formulated an inference-time behavior-shaping framework for pretrained autoregressive driving models, repeatedly selecting native short-horizon action tokens to steer CAV behavior without updating model weights.
- Built and validated a closed-loop mixed-autonomy platform integrating SMART/CAT-K, MetaDrive, and reactive IDM/MOBIL traffic, with matched counterfactual rollouts and action-execution audits.
- Established repeatable CAV control authority from native model choices across four traffic setups, with all paired continuations preserving directional ordering and 12.7–33.0 m separation after 7 s.
- Demonstrated CAV-to-traffic transmission across six geometry–seed cases, linking follower observations to control and speed changes, and laying groundwork for consequence-aware multi-CAV control.

Guangdong Provincial Special Fund for Science and Technology Innovation Strategy 07/2024-04/2026

Title: *Cooperative Game-Based Network Traffic Assignment Mechanisms for Enhancing Urban Transportation Efficiency* | Grant No. PDJH2024A289 | Project Leader | Advisor: Dr. Hu Lei

- Authored a systematic review on dynamic traffic assignment (DTA), synthesizing insights from 120+ studies on multi-class vehicle modeling, nonlinear equilibrium evolution, and solution algorithms^[J1].
- Developed a VI-based DTA model that unifies UE–SO equilibria into a tunable spectrum, by decoupling CAV penetration into automation levels (AL) and connectivity levels (CL) as evolution variables.
- Formulated and evaluated an adaptive mirror–projection algorithm with active-set warm starts for DUE, DSO, and UE–SO mixed equilibria in a node-based CTM framework under AL–CL heterogeneity.
- Directed team task delegation, milestone tracking, and mentoring of lower year members; coordinate regular progress meetings to align efforts and ensure on-schedule research deliverables.

Guangdong Provincial College Student Innovation & Entrepreneur Training Program 11/2024-02/2026

Title: *Intelligent Recognition of Urban Traffic Equilibrium States under Air-Ground Collaborative Monitoring* | Project Co-Leader | Advisor: Prof. Xinhai Xia

- Integrating heterogeneous data sources from roadside sensors, probe vehicles, and UAVs to construct high-dimensional traffic flow tensors, supporting accurate traffic equilibrium state recognition.
- Formulated three representative real-world data loss structures—element-wise, fiber-wise, and mixed—within urban traffic flow tensors, serving as a rigorous testbed for refining data recovery algorithms.
- Developed an adaptive truncated Schatten norm-based tensor completion model to effectively fill gaps in traffic datasets with complex missing patterns, improving MAPE by 10.6% and RMSE by 6.1%^[C3].
- Facilitated industry-academia partnership for traffic survey equipment and training, guided junior students in fieldwork to create datasets for analysing traffic flow dynamics and defining equilibrium states.

High-Quality Development Project for Hundred Counties, Thousand Towns, Myriad Villages 07/2024

Title: *Investigation on Traffic Safety and Planning in Kaiping City* | Project Leader | Advisor: Dr. Hu Lei

- Collaborated with the Municipal Transportation Bureau to analyse traffic accident and violation data, identifying crash hotspots and associated hazard types, and designing two targeted questionnaires.
- Led a seven-person team to conduct on-site surveys, interviews, inspections, and drone photography, collecting data on traffic flow, vehicle management, pedestrian movement, and infrastructure.
- Utilised SPSS and LLM tools for analysis of survey data and interview transcripts, uncovering key issues and generating actionable insights to inform improvements in urban traffic safety and planning.
- Authored two papers—one on traffic safety behaviour, attitude and cognition relationship using Structural Equation Modeling^[J2], and another on the city's traffic governance assessment via Grey Relational and Fuzzy Evaluation^[J3]; a synthesized report was later adopted by the Municipal Transportation Bureau.

INTERNSHIP

Guangzhou Xueshujia Software Technology Co., Ltd.

01/2025-05/2025

Simulation Assistant Engineer | Mentor: Dr. Wenchao Shen

- Deployed a multi-agent reinforcement learning module for CAV control in the company's traffic simulator, enhancing its capability to model and visualise V2X-based highway shock-wave propagation.
- Benchmarked the simulator against leading platforms such as SUMO using identical real-world trajectory datasets, diagnosing fidelity gaps and recommending optimisations to enhance competitiveness.
- Surveyed major traffic-related competitions and conferences over the past five years, analysing emerging trends in the field and discovering potential opportunities for industry–academia collaboration.

AWARDS & HONOURS

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| ➤ IEEE ITSS Travel Grant, The 29th ITSC Conference, ITSS | 2026 |
| ➤ Best Poster Award, The 25th CICTP Conference, COTA | 2025 |
| ➤ Meritorious Winner, Interdisciplinary Contest in Modeling (ICM), COMAP | 2025 |
| ➤ Provincial Third Prize, The 18th "Challenge Cup" National Students' Academic Works Competition | 2025 |
| ➤ Yang Hanlin Scholarship (Top 0.2%, 3/1210), Guangzhou Maritime University | 2025 |
| ➤ Provincial Third Prize, China Undergraduate Mathematical Contest in Modeling | 2024 |
| ➤ Science and Technology Innovation Model Scholarship, Guangzhou Maritime University | 2024 |

SKILLS & LANGUAGES

Technical Tools: Python, MATLAB, SUMO, LaTeX, SPSS, Photoshop, Illustrator

Languages: English (IELTS Academic 6.5), Chinese (native, Mandarin & Cantonese)